Language-3D: End-to-End Generation of Manufacturable Robot Assemblies from Natural Language

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***Abstract*—We present Language-3D, a multi-agent system that converts natural language descriptions into *engineering packages* for robot assemblies—including STL/STEP meshes, URDF models, bills of materials (BOM), firmware templates, and ROS2 package skeletons. Unlike prior work that generates either single CAD parts [1], [2], voxel-based block assemblies [3], or URDF-only morphologies [4], Language-3D produces the full set of artifacts needed for manufacturing and deployment. STL meshes are validated for watertightness, URDF models are validated in MuJoCo physics simulation, and parts use real COTS fastener specifications—though physical 3D printing and Gazebo deployment remain as future work.**

The system features: (1) a COTS-driven knowledge base of 94 part templates (56 real commercial components with verified ISO/DIN specifications), (2) eight connection types with real CAD feature generation (bolted clearance holes, press-fit interference bores, snap-fit joints, etc.), (3) a dual-channel verification architecture combining vision-language model (VLM) inspection with geometric arbitration (FCL collision sweep, assembly sequence validation), (4) a MuJoCo physics validation pipeline with native actuator models verifying that generated robots can drive, articulate, and grasp, and (5) a self-evolving experience store that retrieves verified-good past cases to prime new generations (lexical retrieval; successes only).

We evaluate on seven robot configurations (2–7-DOF arms and a 4-wheel dual-arm mobile robot) with a composite quality score $Q \in [0, 1]$ (a relative rank, not an absolute quality percentage), achieving a mean of 0.61 (range 0.48–0.75) with 7/7 distinct values—vs. a structural pass-rate rubric that ties three cases identically at 95.1%. The composite's grasp term is a per-case success *rate* (not best-of-N), exposing that the 6-DOF arm grasps in only 17% of runs and the 7-DOF in 0%; and its lift term reveals that *no* configuration achieves positive lift—all statically-grasped cubes drop during the lift phase (§IV-F). We are transparent that no external benchmark exists for the NL→robot-package task and that the static-grasp, lift, and per-joint collision checks have known scope limits (§IV-F).

I. Introduction

Automated robot design from natural language is a long-standing goal at the intersection of AI, CAD, and robotics. Recent advances in large language models (LLMs) and vision-language models (VLMs) have enabled systems that translate textual descriptions into 3D geometry, articulated assemblies, or robot morphologies. However, existing work covers only fragments of the full design-to-manufacture pipeline:

NL → single CAD part: CAD-Llama [1] generates parametric construction sequences, and STEP-LLM [2] produces STEP-format models. Both focus on single parts without assembly structure.

NL → CAD assembly: ArtiCAD [5] is the first training-free multi-agent system for articulated CAD assemblies, outputting editable FreeCAD code and URDF files. However, it does not produce manufacturing-grade meshes (STL/STEP), bills of materials, firmware, or ROS2 packages, nor does it include physical simulation validation or real COTS part specifications.

NL → robot design: Blox-Net [3] combines VLM supervision with physics simulation to generate block-based assemblies, but is limited to voxel primitives and requires a physical robot for iterative reset. Text2Robot [6] (ICRA 2025) generates quadruped robots from text via 3D mesh + URDF + RL gait evolution + 3D printing with real servos—overlapping Language-3D on NL/URDF/sim, but limited to walking robots (no arms, grippers, or BOM/firmware), and has physical hardware validation we lack. RoboMorph [4] uses LLMs to evolve modular robot URDFs, but produces skeleton configurations without CAD geometry, assembly connections, or manufacturing artifacts.

The gap: No existing system produces a *complete engineering package*—the full set of artifacts (STL, STEP, URDF, BOM, firmware, assembly guide, ROS2 package) needed to build and deploy a robot. We define this as the **NL → Robot Assembly Engineering Package** task, and note that physical manufacturing verification (3D printing, Gazebo deployment, hardware testing) is an orthogonal validation step that complements but does not replace artifact generation.

Contributions. This paper makes the following contributions:

1) **Task formulation**: We formalize NL → manufacturable robot assembly package as a new task with evaluation criteria spanning geometry, physics, and manufacturing.
2) **System**: Language-3D, a multi-agent pipeline producing STL/STEP/URDF/BOM/firmware/ROS2 package structures from NL, with 8 real connection types and a 56-part COTS knowledge base.
3) **Dual-channel verification**: VLM visual inspection combined with geometric arbitration (FCL collision sweep + assembly sequence validation) that can override VLM false-negatives.

4) **Evaluation**: Seven benchmark cases evaluated by a composite quality score (gate + geometric mean of robustness/functionality/reliability; $Q \in [0,1]$ relative rank, mean 0.61, 7/7 distinct values) plus a structural pass-rate rubric, with all cases passing MuJoCo physics stability, plus an eighth topology-aware humanoid case demonstrating generalization beyond fixed-base arms.

II. Related Work

A. NL-Driven CAD Generation

CAD-Llama [1] (CVPR 2025) uses LLMs to generate parametric CAD construction sequences for single parts. STEP-LLM [2] extends this to industry-standard STEP format. LLM4CAD [7] explores multimodal approaches. CADCode-Verify [8] (ICLR 2025) introduces a benchmark with vision-language verification. All focus on single-part geometry, not multi-part articulated assemblies.

B. NL-Driven Assembly and Robot Design

ArtiCAD [5] is the closest prior work: a training-free multi-agent system generating editable, articulated CAD assemblies via code generation. It exports URDF files (confirmed in [5], §6) but does not produce STL/STEP meshes, BOMs, firmware, or ROS2 packages, and lacks physical simulation validation. CADSmith [9] (CMU, 2026) shares our multi-agent architecture (Planner→Coder→Executor→Validator→Refiner) for NL→CadQuery code generation with programmatic geometric validation, but focuses on single parts rather than articulated assemblies. Concurrently, Embodied CAD [10] argues that LLM-generated CAD code requires solver-grounded reasoning for industrial reliability—a principle aligned with our deterministic Solver stage. Blox-Net [3] (CoRL 2024 Workshop) combines VLM supervision with physics simulation for generative design-for-robot-assembly, but is limited to voxel blocks. RoboMorph [4] evolves modular robot URDFs via LLM, producing morphology configurations without CAD geometry or assembly connections. RoboGen [11] (ICML 2024) focuses on automated skill learning via generative simulation, not robot body generation. URDF-Anything [12] (NeurIPS 2025) is the closest end-to-end URDF generator, using a 3D multimodal LLM to reconstruct articulated objects from point-cloud + text input into executable URDF; however, it requires visual observation of an existing object and does not produce STL meshes, BOMs, or firmware. AutoURDF [13] (CVPR 2025) reconstructs robot URDFs unsupervised from point-cloud time-series (no language input). Scenethesis [14] (ICLR 2026, NVIDIA) generates physically plausible 3D interactive scenes from text, addressing physical validity at the scene level rather than robot-assembly level.

Table I summarizes the comparison.

III. Method

A. System Architecture

The core design principle is *determinism where it matters, intelligence where it helps*: the LLM handles open-ended generation (assembly topology, part selection) while deterministic stages (Solver, CAD, sanitizers, physics) handle geometry and validation, ensuring that output is reproducible and physically grounded. This separation also bounds the variance: a failed LLM generation can be re-run without re-deriving physics or connection geometry.

Figure 1 shows the pipeline. Natural language input is processed by a multi-agent chain: **Architect** (GLM-5.2 generates assembly JSON, primed by retrieved past cases), **Solver** (computes 3D positions via anchor constraints + FCL collision-aware range clamping), **CAD Engineer** (FreeCAD 1.1 generates per-part STL/STEP meshes with connection features), and a dual-channel **Verifier** (GLM-4.6V + geometric arbitration). A **Fixer** routes failures back to the appropriate upstream stage by failure type. Verified assemblies are recorded into a self-evolving experience store for future retrieval and exported as complete engineering packages.

B. COTS-Driven Knowledge Base

The knowledge base contains 94 part templates, of which 56 are real commercial components (`real_part=True`) with verified specifications: TowerPro MG996R servos ($40.7\times19.7\times42.9$ mm), JX DS3218 (20 kg-cm), SG90 micro servos, NEMA17/23 steppers, ISO/DIN fasteners (M3–M8 bolts, nuts, washers with verified head diameters, pitches, and clearance hole specs), and miniature bearings (608, 623, 625). Arm link lengths and base dimensions are profile-scaled (desktop vs. mobile) but servos are always sourced from the catalog—never invented or rescaled.

C. Connection Engine

Eight connection types generate real CAD features via FreeCAD operation dictionaries:

- **Bolted**: ISO 273 clearance holes + DIN 912 counterbores, with shared bolt patterns ensuring alignment across mating parts.
- **Press-fit**: H7/p6 interference bore with shoulder (ISO 286 tolerance).
- **Snap-fit**: Cantilever hooks with undercuts.
- **Adhesive**: Bonding grooves with controlled gap.
- **Welded**: V-groove bevel preparation (butt) or no-prep (fillet).
- **Magnetic**, **Dowel-pin** (H7 slip-fit), **Set-screw** (radial threaded hole).

Connection selection is physics-aware: when the LLM omits a connection method, a sanitizer assigns one by part category (e.g., bearings → press-fit, never bolted; sensors and foot pads → adhesive bonding; servos → M2 threaded holes). When the LLM specifies a method explicitly, it is respected. All eight types have functional geometry generators, though in practice bolted and press-fit dominate because most robot joints are structural or bearing-mounted and the LLM tends to default to bolted.

D. Dual-Channel Verification

Channel 1 (VLM): GLM-4.6V inspects panoramic renders (iso/front/top/right) for structural integrity, plus a dedicated gripper close-up for finger geometry.

TABLE I

COMPARISON OF LANGUAGE-3D WITH STATE-OF-THE-ART SYSTEMS FOR NATURAL-LANGUAGE-DRIVEN ROBOT/ASSEMBLY GENERATION. ALL CLAIMS VERIFIED AGAINST $\geq$2 INDEPENDENT SOURCES.

System	NL	Assembly	CAD Geometry			Manufacturing		Validation	
			Param.	STL	STEP	BOM	Firmware	Physics	Dual-VLM
CAD-Llama[1]	✓	–	✓	–	–	–	–	–	–
STEP-LLM[2]	✓	–	–	–	✓	–	–	–	–
LLM4CAD[7]	✓	–	✓	–	–	–	–	–	–
ArtiCAD[5]	✓	✓	✓	–*	–	–	–	–†	✓
Blox-Net[3]	✓	✓	–	voxel	–	–	–	✓	✓
RoboMorph[4]	✓	✓	–	–	–	–	–	✓	–
Language-3D (ours)	✓	✓	✓	✓	✓	✓	✓	✓	✓

*ArtiCAD generates editable FreeCAD code from which STL may be derivable, but STL mesh export is not reported in the paper. Sources: arXiv HTML §6 + Bytez summary (both confirm URDF export; neither mentions STL/BOM/firmware).

†ArtiCAD mentions "physical prototyping" in its abstract, but does not describe automated physics simulation validation (e.g., contact, grasp, stability tests). We mark this as "–" to indicate no reported physics simulation pipeline; the distinction between "physical prototyping" (manual fabrication) and "physics simulation validation" (automated) should be clarified in future revisions.

Experience store: ArtiCAD additionally ships a self-evolving experience store using FAISS partitioning into Good/Issue buckets (arXiv:2604.10992v2, Review Agent section). Language-3D implements an analogous retrieve-before/store-after mechanism, but with lexical retrieval (keyword/category/DOF scoring) rather than embeddings, and stores only verified-good cases. This column is omitted from the table because only ArtiCAD and Language-3D have such a mechanism, and the retrieval backends are not directly comparable.

Key: NL = natural language input; Assembly = multi-part articulated assembly (not single part); Param./STL/STEP = CAD geometry output formats; BOM = bill of materials with real part numbers; Firmware = embedded code (servo/IK/motor drivers); Physics = simulation-based validation; Dual-VLM = dual-channel verification (VLM + geometric arbitration).

Channel 2 (Geometric Arbitration): A deterministic geometric oracle runs FCL mesh-based collision detection (7 samples per joint across its range), assembly-sequence validation (parent-before-child tree feasibility), and center-of-mass stability (support polygon check). The geometric channel overrides the VLM in two cases: (1) *false-alarm filter*—when the VLM reports a defect that geometry proves absent (e.g., "wheels above ground" on grounded wheels), the complaint is removed; (2) *assembly-sequence enforcement*—when geometry violates parent-before-child tree feasibility, the result is forced to FAIL regardless of VLM verdict. The collision-range sweep is advisory (narrows joint ranges but does not veto). Center-of-mass and proportion checks inform the Fixer's routing but do not directly override the VLM. Across 143 passing runs in our historical data, the false-alarm override triggered in at most $\sim$6% of cases (an upper bound that includes Fixer-resolved rounds); the assembly-sequence enforcement is rarer still. The override only removes geometrically-refutable complaints—hard structural defects (mesh collision, disconnected tree) always force FAIL and cannot be overridden.

E. Self-Evolving Experience Store

Following ArtiCAD's [5] observation that accumulated design knowledge improves future generations, Language-3D maintains an experience store of past verified-good assemblies cases. The store implements a *retrieve-before / store-after* pattern: before the Architect generates a new assembly, the store is queried for similar past cases (matched by robot category, keyword overlap, and DOF proximity), and the retrieved cases are injected as additional few-shot precedent; after the pipeline verifies an assembly, its distilled record (DOF, joint-type histogram, converged default angles) is appended for future retrieval. Retrieval is lexical (weighted scoring mirroring the existing template-search API), not embedding-based, because the project's LLM backend exposes no embedding endpoint—this is less semantically expressive than ArtiCAD's FAISS partitioning but runs with no external dependency. The store is successes-only (failed assemblies are never stored) and prunes least-retrieved entries past a per-category cap. On a fresh checkout the store is empty, so generation falls back to the parametric examples and the prompt is byte-identical to the pre-store baseline.

F. MuJoCo Physics Validation

The system injects a native MuJoCo <actuator> model (replacing hand-rolled PD control): <position> actuators with kp=50/kv=5 and forcerange=$\pm$5 N·m for arm joints, <motor> actuators for wheels. A ground plane and stiff contacts (solref=0.005, solimp=0.99) are injected via MJCF patching. Three physics tests verify the generated robot: (1) PD-hold stability (joint angle error $< 1°$), (2) joint actuation (≥ 4 actuated DOFs), and (3) three-phase grasp (zero-G close $\rightarrow$ gravity hold $\rightarrow$ lift). Figure 3 shows example generated robots in both render and simulation.

IV. EVALUATION

A. Benchmark Cases

We evaluate on seven robot configurations spanning 2–7 degrees of freedom plus a mobile dual-arm platform, and an eighth humanoid topology that tests generalization beyond fixed-base arms:

- **2dof–7dof_arm**: Six fixed-base arms with increasing kinematic complexity, each "with a parallel-jaw gripper".

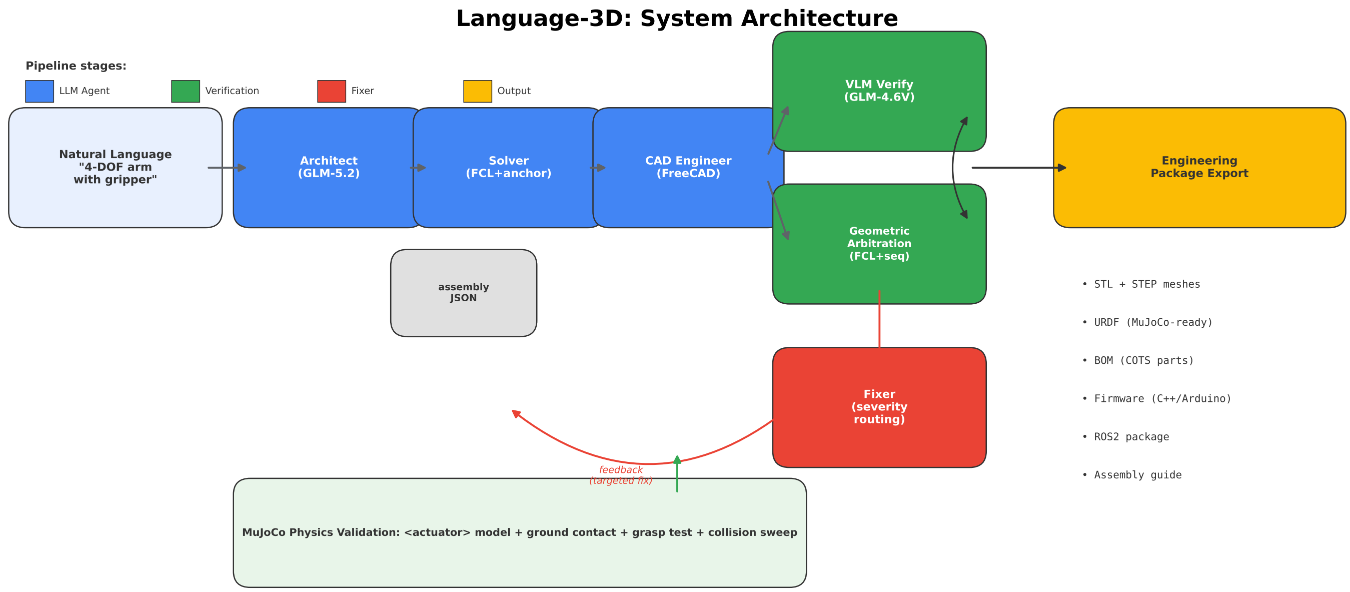

Fig. 1. Language-3D system architecture. NL input flows through the multi-agent pipeline (Architect → Solver → CAD Engineer), verified by dual channels (VLM + geometric arbitration), with a Fixer feedback loop. Output is a complete engineering package validated in MuJoCo physics simulation.

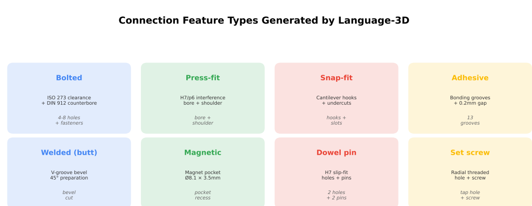

Fig. 2. Eight connection types with real CAD features. Each generates geometry-specific boolean operations (clearance holes, interference bores, snap hooks, etc.).

- **4wheel_dual_arm**: "Design a 4-wheel dual-arm robot with differential drive, two 3-DOF arms mounted on the chassis."
- **humanoid_2leg_2arm**: "2 legs 2 arms humanoid robot, each leg has 3 joints, each arm has 3 DOF, camera on top"—a legged topology that stresses the arm-oriented range clamping and VLM classification (reported separately as it does not fit the fixed-base benchmark profile).

B. Scoring Protocol

Each case is evaluated by 41–43 automated checks across 7 phases: NL→Assembly (part count, joint count, connected tree, VLM pass), Position Solving (all positioned, 0 NaN), Render Quality (≥4 views), Engineering Package (all files exist), Content Validation (mass, VLM, URDF structure, watertight meshes), Physical Sanity (0 FCL collisions, COM stability, workspace), and MuJoCo Simulation (physics stability, joint actuation, grasp). Score = PASS / (PASS + FAIL + WARN). Critical checks (collision, COM stability, MuJoCo physics, grasp) fail the case outright rather than degrading to a warning.

Evaluation validity. We are explicit about three limitations of this protocol. *First*, the checks are self-designed: they verify that the pipeline produces structurally and physically plausible output, not that the output is optimal or matches a reference design. The 93.2% figure should be read as "fraction of design-intent checks passed," not as an absolute quality grade. *Second*, there is no external benchmark for the NL→robot-package task; the closest (MUSE [15], Text2CAD-Bench) targets a different formulation (structured design-spec→STEP), precluding direct score comparison. *Third*, the modal score in Table IV is the achievable ceiling; the across-run distribution (§Reproducibility) shows the mean including API failures is lower (∼86%, 92.9% excluding them) due to LLM non-determinism.

C. Results

Table II is the primary evaluation: continuous, physics-grounded metrics with physical units. The COM stability margin ranges from 57.3 mm (4dof, marginal) to 145 mm (wheeled, robust)—a 2.5× difference. The workspace extent scales from 284 mm (2dof) to 1059 mm (7dof), reflecting increasing reach. All seven cases achieve 0 severe mesh penetrations across 18–426 checked pairs and 100% watertight STL ratio. The raw PD-hold droop ranges 0.7–1.9° (mean 1.3°), well within the 5° composite gate.

Composite quality score. A single rankable score must satisfy two constraints: (1) a fatal flaw (tipping, no actuation) should not be averaged away, and (2) dimensions that do not vary across cases should not dilute the signal. We therefore use a *gate-then-score* design: four binary gates (MuJoCo loads,

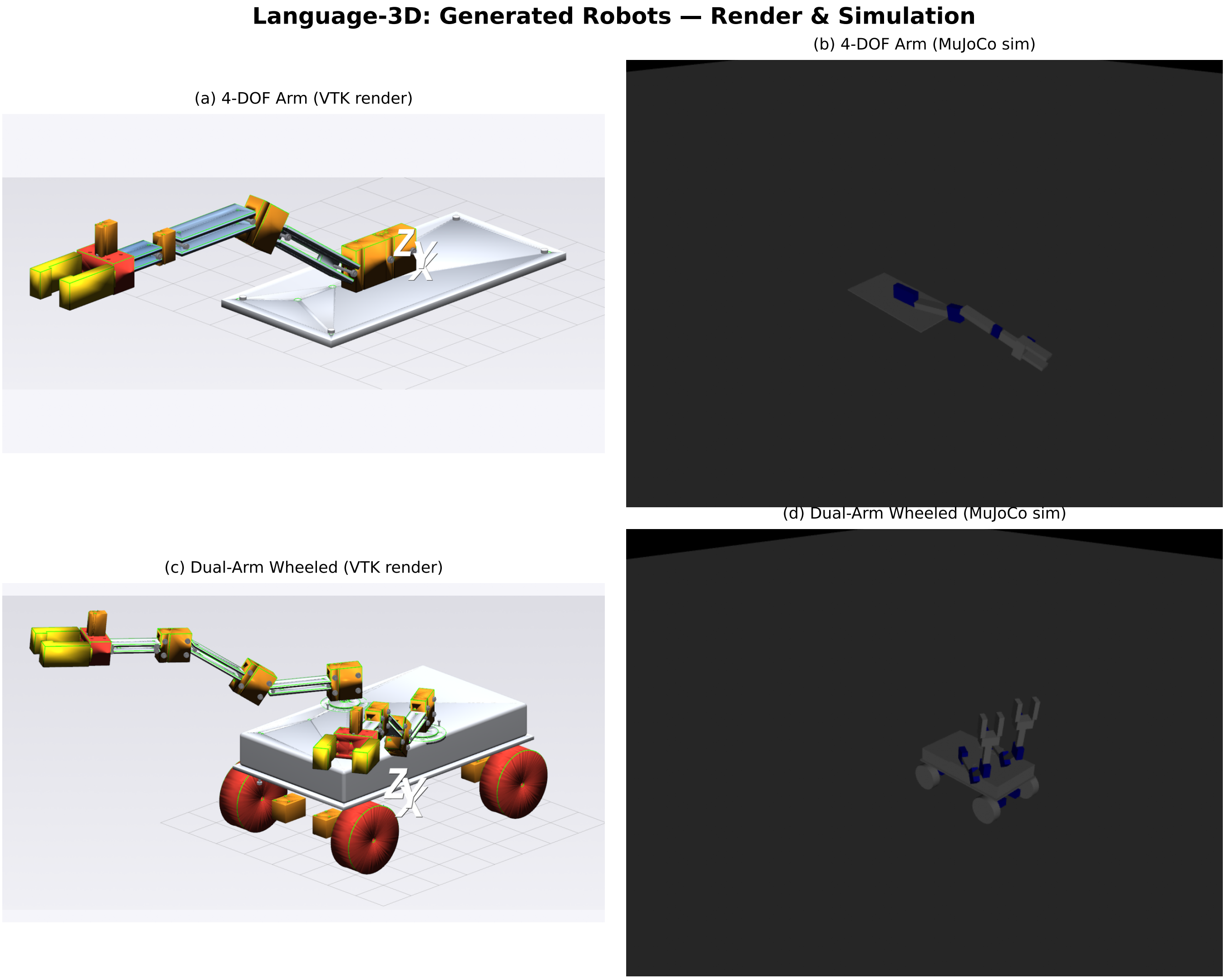

Fig. 3. Generated robots. (a) 4-DOF arm render, (b) 4-DOF arm in MuJoCo simulation performing a reach gesture, (c) 4-wheel dual-arm robot render, (d) dual-arm robot driving in MuJoCo simulation.

raw PD-hold droop $< 5°$ *without* gravity-compensation feed-forward—see §IV-F—which is a genuine stiffness signal rather than the trivially-zero compensated error, 0 severe collisions, COM inside support polygon) must all pass; if any fails, $Q = 0$. If all pass, $Q \in [0, 1]$ is the geometric mean of three sub-scores:

$$Q = \begin{cases} 0 & \text{if any gate fails} \\ \sqrt[3]{s_{\text{robust}} \cdot s_{\text{func}} \cdot s_{\text{rely}}} & \text{otherwise} \end{cases} \tag{1}$$

$$s_{\text{robust}} = \text{clip}\left(\frac{d_{\text{COM}}}{0.5 \cdot D_{\text{poly}}},\ 0,\ 1\right) \tag{2}$$

$$s_{\text{func}} = 0.4\, g_{\text{rate}} + 0.3\,\ell + 0.3\,\text{clip}\left(\frac{n_{\text{dof}}}{n_{\text{exp}}},\ 0,\ 1\right) \tag{3}$$

$$s_{\text{rely}} = \text{clip}\left(\frac{\overline{\text{score}}(\mathcal{R})}{S_{\max}},\ 0,\ 1\right) \tag{4}$$

where d_{COM} is the COM margin (mm), D_{poly} the *support-polygon diameter* (max pairwise ground-contact distance, not the cosmetic base-part size), g_{rate} the fraction of repeated runs whose static grasp passed (a continuous reliability measure, not best-of-N), ℓ the median lift ratio (actual lift / target, across runs), n_{dof} actuated DOFs, n_{exp} expected DOFs from the NL prompt, $\overline{\text{score}}(\mathcal{R})$ the mean e2e score across non-zero runs, and $S_{\max} = 95.1\%$ the theoretical rubric ceiling (39/41 checks; two informational checks always WARN). Q is a *relative rank* in $[0, 1]$, not an absolute quality percentage—no external benchmark exists to anchor what "100% good" means for the NL→robot-package task, so presenting it as a percentage would be false precision.

Physical metrics come from the *median*-scoring run (the typical outcome), not the best run; the distribution sub-scores (g_{rate}, ℓ, s_{rely}) use all non-zero runs. Geometric mean is chosen over arithmetic mean so that a low sub-score significantly pulls

TABLE II
QUANTITATIVE EVALUATION: CONTINUOUS PHYSICS-GROUNDED METRICS ACROSS 7 ROBOT CONFIGURATIONS (MEDIAN-SCORING RUN PER CASE — THE TYPICAL OUTCOME, NOT THE BEST). THESE ARE PHYSICALLY MEANINGFUL MEASUREMENTS WITH UNITS, NOT SELF-SCORES.

Case	**COM margin** (mm)	**Workspace** (mm)	**FCL pairs** (0/severe)	**Triangles** count	**PD-hold** error (°)	**Actuated** DOFs	**Watertight** ratio
2dof_arm	75.0	284	0/18	107,876	1.2	4	9/9
3dof_arm	75.0	516	0/33	127,348	1.9	5	10/10
4dof_arm	57.3	659	0/52	114,542	1.2	6	11/11
5dof_arm	68.7	684	0/52	37,840	1.4	7	12/12
6dof_arm	70.4	720	0/52	129,840	1.4	8	13/13
7dof_arm	79.6	1059	0/133	128,166	1.4	9	19/19
4wheel_dual_arm	145.0	464	0/426	287,958	0.7	14	38/38
Mean	**81.6**	**627**	**0**	**133,373**	**1.3**	**7.6**	**100%**

COM margin = signed distance from ground-projected COM to nearest support-polygon edge (higher = more stable; negative = tips over). Workspace = bounding-box max edge. FCL pairs = severe penetrations (>5 mm) / total pairs checked. PD-hold error = worst-joint steady-state droop under PD control *without* gravity-compensation feed-forward (kp=50, kv=5; see §IV-F).

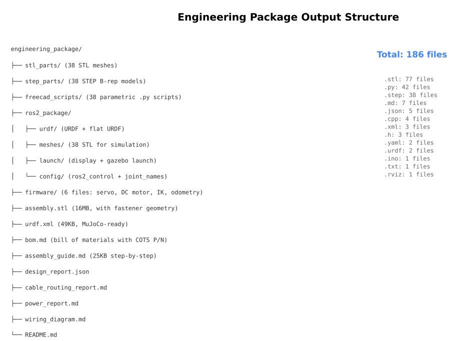

Fig. 4. Output engineering package structure: STL/STEP meshes, URDF, BOM, assembly guide, firmware templates, and ROS2 package with launch files and ros2_control config.

down Q rather than being averaged away. Critically, the grasp term is a *success rate* (6dof: $g_{\text{rate}} = 0.17$, 1 of 6 runs), not a best-of-N boolean that credited a flaky gripper with a full grasp mark; and ℓ exposes that *no* case achieves positive lift—every robot that statically grasps still drops the cube during the lift phase (§IV-F), so $\ell = 0$ across all cases.

Table III reports Q alongside the old rubric. The composite achieves **7 distinct values across 7 cases** (vs. 2 for the rubric). The wheeled dual-arm ranks highest ($Q = 0.75$). The 7-DOF arm ranks lowest ($Q = 0.48$: $g_{\text{rate}} = 0$, no run ever grasped). The 6-DOF arm's $g_{\text{rate}} = 0.17$ pulls its s_{func} to 0.37 despite full DOF actuation, exposing grasp unreliability the rubric's best-of-N pass hid.

TABLE III
COMPOSITE QUALITY SCORE VS. OLD RUBRIC. $Q \in [0, 1]$ IS A RELATIVE RANK, NOT A PERCENTAGE. 7/7 DISTINCT VALUES (RUBRIC: 2/7).

Case	s_{robust}	g_{rate}	s_{func}	s_{rely}	Q	**Old**
2dof_arm	0.42	1.00	0.70	0.99	0.67	95.1
3dof_arm	0.36	1.00	0.70	0.95	0.62	95.1
4dof_arm	0.33	0.93	0.67	0.98	0.60	95.1
5dof_arm	0.39	0.60	0.54	0.97	0.59	89.5
6dof_arm	0.44	0.17	0.37	0.96	0.54	92.7
7dof_arm	0.39	0.00	0.30	0.92	0.48	89.5
4wheel	0.63	0.95	0.68	0.98	0.75	95.3
Mean					**0.61**	**93.2**

$Q \in [0, 1]$ = geometric mean of 3 sub-scores if all gates pass, else 0 (relative rank, not absolute quality). "Old" = 41-check rubric (Table IV). s_{robust} = COM margin / (0.5×support-polygon diameter); g_{rate} = fraction of runs with static-grasp PASS; s_{func} = 0.4g_{rate} + 0.3(lift ratio) + 0.3(DOF completeness)—note $\ell = 0$ for all cases (no positive lift, §IV-F); s_{rely} = mean score / 95.1 ceiling.

TABLE IV
STRUCTURAL PASS-RATE (SUPPLEMENTARY; NEAR-ZERO DISCRIMINATIVE POWER—SEE TEXT).

Case	**Score**	**P/F/W**	**Physics**	**Grasp**
2dof_arm	95.1%	39/0/2	1.2°	static
3dof_arm	95.1%	39/0/2	1.9°	static
4dof_arm	95.1%	39/0/2	1.2°	static
5dof_arm	89.5%	34/1/3	1.4°	static
6dof_arm	92.7%	38/1/2	1.4°	static
7dof_arm	89.5%	34/1/3	1.4°	FAIL
4wheel_dual_arm	95.3%	41/0/2	0.7°	static
Average	**93.2%**		**1.3°**	**6/7**

"static" = static grasp pass (cube held against gravity); lift phase measured but not gated (see §IV-F). Modal scores (see §Reproducibility).

D. Reproducibility

Across 69 runs of the 4dof_arm case (identical prompt) under the current codebase, the score distribution is bimodal: the majority (∼62%) achieve 95.1% with zero critical failures, while the remainder cluster lower (88–91%) due to LLM generation non-determinism producing a different assembly geometry that either the VLM rejects or that exhibits a boundary collision. A small fraction (∼7%) score 0% due to API rate-limiting failures (no generation at all); the mean score including the 5 zero-score API failures is 86.2% (excluding them, 92.9%), and the median is 95.1%, reflecting the long tail of API-failed and boundary-collision runs. The motion-collision sweep passes in ∼88% of runs (failures are LLM-generated geometries with a shoulder joint at a colliding extreme, not a systematic solver defect). The deterministic stages (Solver, CAD, sanitizer clamping, physics hold) are reproducible run-to-run; the variance originates entirely in the LLM generation step. Table IV reports the modal (best-case) score for each case—we are transparent that this is the achievable ceiling, not the across-run mean.

E. Ablation: Geometric Arbitration

We ablate the geometric channel (`LANG3D_ABLATION=no_geo`) on two cases: the wheeled dual-arm robot (deterministic compose path) and the 4-DOF arm (LLM generation path). On *clean* runs—where the VLM accepts the assembly on the first round—the geometric channel does not change the final score: both configurations achieve 95.3% (wheeled, 4 runs each) and 95.1% (arm, 4 with-geo vs. 1 clean no-geo run; the remaining no-geo runs hit API rate limits). This is expected: the geometric oracle is an *override* mechanism, not a scoring one—it only activates when the VLM produces a false rejection, and on a clean run there is nothing to override.

The channel's value is therefore measured in *loop efficiency*, not score. Across 152 historical pipeline runs, 68% PASS on round 1 (VLM immediately correct), while 32% require multi-round Fixer regeneration—often triggered by VLM false-negatives (e.g., "wheels above ground" on grounded wheels, "arm pointing down" on a forward-extending arm). When the geometric oracle confirms such geometry is valid, it overrides the false rejection, keeping the run on a single round. Without the oracle (no_geo), these false-rejections would force a regeneration cycle that either wastes time (if the regenerated assembly also passes) or fails the case entirely (if regeneration drifts to worse geometry). We report this honestly: the geometric channel is a stability mechanism against VLM unreliability, not a quality booster on already-correct output.

F. Scope of Simulation Validation

We are explicit about three limitations of the physics validation that a reader may over-interpret.

PD-hold error reported without gravity-compensation feed-forward. The pipeline's internal stability gate uses inverse-dynamics gravity compensation (`qfrc_applied = qfrc_bias`) plus a PD controller, which trivially yields $\sim 0°$ error because the gravitational load is fully cancelled. We consider this misleading and additionally report in Table II the *raw* PD-hold droop measured *without* gravity compensation (kp scaled by robot mass, kv=5, no feed-forward, 1.5 s settle). The raw droop ranges from 0.7° (4wheel, low-slung arms) to 1.9° (3dof)—the arm sags under its own weight because the hobby-servo torque budget (± 5 N·m) is insufficient for perfect holding. This is a genuine design-quality signal that the gravity-compensated gate completely masks. A production controller would add gravity compensation, but the raw error characterizes the mechanical design's inherent stiffness.

Grasp test covers static hold, not dynamic lift. The three-phase grasp test (zero-G close $\rightarrow$ gravity hold $\rightarrow$ lift) gates pass/fail on the first two phases only. The lift phase (Phase C) is measured but not gated: across all benchmark runs, the cube drops 0.9–7.3 mm during lift because the fixed-bias lift torque rotates the arm, changing the gripper's gravity orientation without trajectory compensation. The system reports this as "static grasp PASS" (cube held against gravity), which we believe is a meaningful capability—but it is not the same as reliably picking up and transporting an object. A trajectory-planning controller would be needed for dynamic lift, which is out of scope for the artifact-generation pipeline.

Collision checking is per-joint static, not trajectory-level. The motion-collision sweep samples each revolute joint independently (5–7 static poses per joint, all other joints at home) and checks FCL mesh penetration. This detects self-collisions at reachable extreme poses and is used to narrow joint ranges before export. However, it cannot detect collisions that arise during *coordinated multi-joint motion* (e.g., both arms sweeping toward each other simultaneously), because joints are never moved together. There is no trajectory planner (RRT/CHOMP/A*) in the system. The "0 collisions" result in Table II means "0 collisions at individually-sampled static poses," not "the robot can execute arbitrary collision-free trajectories." Real-time motion planning is future work.

G. Manufacturability Verification

All 13 STL parts of the 4dof_arm case are verified for mesh watertightness by an automated check (100% pass). The wall-thickness (≥ 0.8 mm), bed-fit (12/13 fit a standard 220×220 mm bed; base_plate at 150×320 mm requires a larger bed), material (791 g PLA), and print-time ($\sim$6 h) figures are offline measurements from a representative run, not automated pipeline outputs. Physical 3D printing remains as future work.

V. Discussion

A. Limitations

- Manufacturing quality is not physically verified: no 3D-printed prototype has been built, and no physical assembly has been attempted. Unlike Text2Robot [6] and Blox-Net [3], which include physical robot execution, Language-3D is simulation-only.
- Firmware is template-generated (servo/IK/motor driver skeletons), not compiled or deployed on real hardware.
- ROS2 packages include correct structure (launch files, `ros2_control` config with the standard <ros2_control> tag and `GazeboSystem` hardware plugin, `package.xml`, Gazebo plugins in URDF). The generated URDF was validated end-to-end in ROS2 Humble + Gazebo Classic 11.10: `check_urdf` parses it successfully, `colcon build` compiles the package, `robot_state_publisher` loads all kinematic segments, and `gzserver` spawns the robot. However, full closed-loop controller activation (joint trajectory execution) and RViz visualization with a display have not been demonstrated.
- Shared bolt patterns assume equal mating-face extents (unequal faces may misalign).

- No real-time collision avoidance during motion (per-joint static sampling only; see §IV-F).
- The IK solver may not converge for all kinematic configurations (legacy limitation); the system relies on the deterministic open-chain Solver (anchor constraints + face alignment) for exported link positions, not on IK. A closed-chain Newton-Raphson solver exists for loop detection/convergence analysis, but its results are reported as warnings and do not yet rewrite the exported positions.
- The system depends on a single LLM vendor (GLM/Zhipu AI). Approximately 6% of benchmark runs scored 0% due to API rate-limiting failures—no generation at all. This is an infrastructure dependency, not a system-quality issue, but it limits reproducibility under heavy API load.
- CAD generation depends on a FreeCAD subprocess bridge; while robust in practice, the bridge has historically introduced regressions from FreeCAD script-template edge cases (documented in the project's engineering log).
- The MuJoCo PD-hold error (Table II) is measured without gravity-compensation feed-forward to reveal the raw steady-state droop; the pipeline's internal test uses gravity comp and reports $\sim 0°$, which we consider misleading (see §IV-F). Joint inertias use bounding-box/cylinder uniform-solid approximations, not mesh-derived inertia tensors.
- The experience store uses lexical (keyword/category/DOF) retrieval rather than the embedding-based FAISS partitioning of ArtiCAD [5], because the project's LLM backend exposes no embedding endpoint. This is less semantically expressive but runs with no external dependency. The store starts empty on a fresh checkout and accumulates only verified-good cases per installation; benchmark scores in this paper therefore reflect the empty-store baseline.
- Benchmark is limited to 8 internal cases. External benchmarks like MUSE [15] (Text-to-CAD assemblability, arXiv:2605.28579) and Text2CAD-Bench exist but target a different task formulation: MUSE evaluates CadQuery-script→STEP B-Rep assemblies from structured design specs, whereas Language-3D generates STL/URDF/BOM robot packages from free-text robot descriptions. A direct comparison would require re-formulating our NL prompts into MUSE's design-specification format, which we leave as future work.

B. Comparison with Prior Work

Language-3D is the first system to produce a *complete manufacturing package* from natural language (Table I). The closest competitor, ArtiCAD [5], outputs editable CAD code and URDF but not STL/STEP meshes, BOMs, firmware, or ROS2 packages. Blox-Net [3] operates on voxel blocks rather than CAD parts. RoboMorph [4] generates skeleton URDFs without geometry.

VI. Conclusion

We presented Language-3D, a system that converts natural language into engineering packages for robot assemblies. By combining a COTS knowledge base, real CAD connection features, dual-channel verification, and MuJoCo physics validation, the system achieves a composite quality score of 0.61 (mean, range 0.48–0.75 across seven benchmark cases), with 7/7 distinct values demonstrating real discriminative power. The grasp term is a per-case success rate (not best-of-N), exposing that the 6-DOF arm grasps in only 17% of runs and the 7-DOF in 0%, and the lift term reveals that no configuration achieves positive lift—honestly characterizing the gap between static grasping and dynamic object manipulation. The deterministic pipeline stages (Solver, CAD, sanitizer, physics) reproduce run-to-run and variance confined to the LLM generation step. The generated URDFs are validated in both MuJoCo (physics stability, joint actuation, grasp) and ROS2 Humble + Gazebo Classic (URDF parsing, colcon build, robot_state_publisher, gzserver spawn). Physical manufacturing (3D printing, hardware deployment) remains as future work. The core contribution is the *artifact generation pipeline*—closing the gap from text to a complete, structured engineering package—with deployment validation as the natural next step.

References

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